

Work

- The transfer of energy to or from a system through the application of a force over a displacement.
- Work tells us how much a force or combination of forces changes the energy of a system.
- Work is the bridge between force (a vector) and energy (a scalar).

$$W = Fd \cos \theta$$

- F: force (N)
- d : displacement (m)
- θ : angle between force and displacement

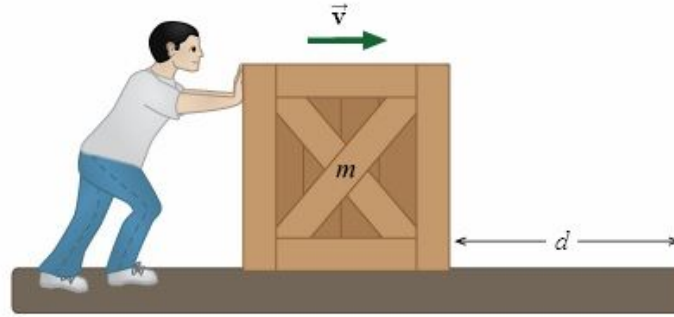
Units of Work

- SI System: Joule (N m)
 - 1 Joule of work is done when 1 N acts on a body moving it a distance of 1 meter
- British System: foot-pound
 - (not used in Physics 1)
- cgs System: erg (dyne-cm)
 - (not used in Physics 1)
- Atomic Level: electron-Volt (eV)

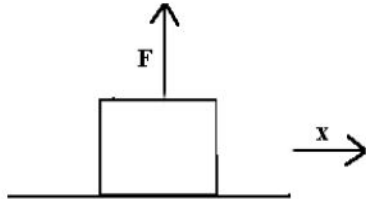
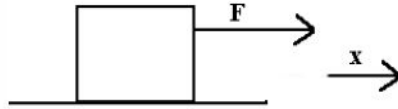
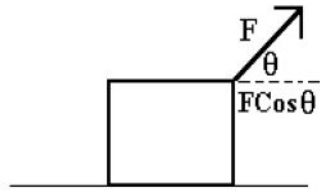
Force and direction of motion both matter in defining work!

- There is no work done by a force if it causes no displacement.
- Forces can do positive, negative, or zero work.

□ When a box is pushed on a flat floor, for example...



- The normal force and gravity do no work, since they are perpendicular to the direction of motion.
- The person pushing the box does positive work, since he is pushing in the direction of motion.
- Friction does negative work, since it points opposite the direction of motion.



The **VERTICAL** component of the force **DOES NOT** cause the block to move the right. The energy imparted to the box is evident by its motion to the right. Therefore **ONLY** the **HORIZONTAL COMPONENT** of the force actually creates energy or **WORK**.

When the **FORCE** and **DISPLACEMENT** are in the **SAME DIRECTION** you get a **POSITIVE WORK VALUE**. The **ANGLE** between the force and displacement is **ZERO** degrees. **What happens when you put this in for the COSINE?**

When the **FORCE** and **DISPLACEMENT** are in the **OPPOSITE** direction, yet still on the same axis, you get a **NEGATIVE WORK VALUE**. **This negative doesn't mean the direction!!!!** IT simply means that the force and displacement oppose each other. The **ANGLE** between the force and displacement in this case is **180** degrees. **What happens when you put this in for the COSINE?**

When the **FORCE** and **DISPLACEMENT** are **PERPENDICULAR**, you get **NO WORK!!!** The **ANGLE** between the force and displacement in this case is **90** degrees. **What happens when you put this in for the COSINE?**

Question

- If a man holds a 50 kg box at arms length for 2 hours as he stands still, how much work does he do on the box?
- ZERO

Question

- If a man holds a 50 kg box at arms length for 2 hours as he walks 1 km forward, how much work does he do on the box?
- ZERO

Question

- If a man lifts a 50 kg box to a height of 2.0 meters, how much work does he do on the box?
- 1000 J

Sample problem

Jane uses a vine wrapped around a pulley to lift a 70-kg Tarzan at a constant speed to a tree house 9.0 meters above the ground.

a) How much work does Jane do when she lifts Tarzan?

6300 J

b) How much work does gravity do when Jane lifts Tarzan?

-6300 J

Sample problem

- A droplet of water of mass 50 mg falls at constant speed under the influence of gravity and air resistance. After the drop has fallen 1.0 km, what is the work done by
- a) gravity?
0.500 J
- b) air resistance?
-0.500 J
- c) What is the net work?
0 J

When an object moves at a constant speed the net work done on the object is zero.

Sample problem

A father pulls his child in a little red wagon with constant speed. If the father pulls with a force of 16 N for 10.0 m, and the handle of the wagon is inclined at an angle of 60° above the horizontal, how much work does the father do on the wagon?

80 J

Sample problem

Joe pushes a 10-kg box and slides it across the floor at constant velocity of 3.0 m/s. The coefficient of kinetic friction between the box and floor is 0.50.

a) How much work does Joe do if he pushes the box for 15 meters?

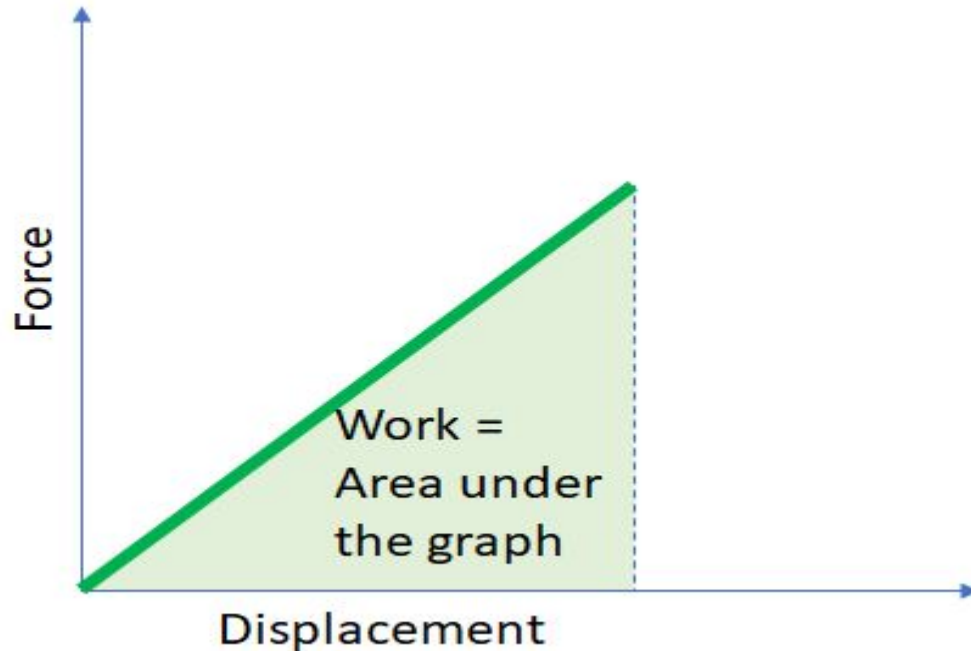
750 J

b) How much work does friction do as Joe pushes the box?

- 750 J

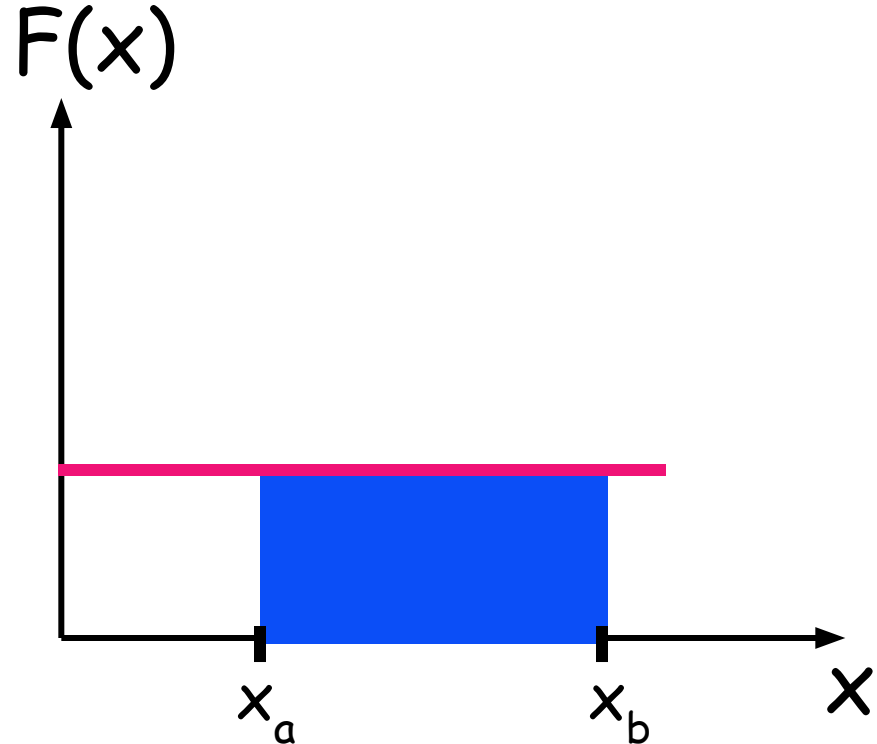
Determining work from a force vs displacement graph

The area under a force vs displacement graph is the work done by the force.

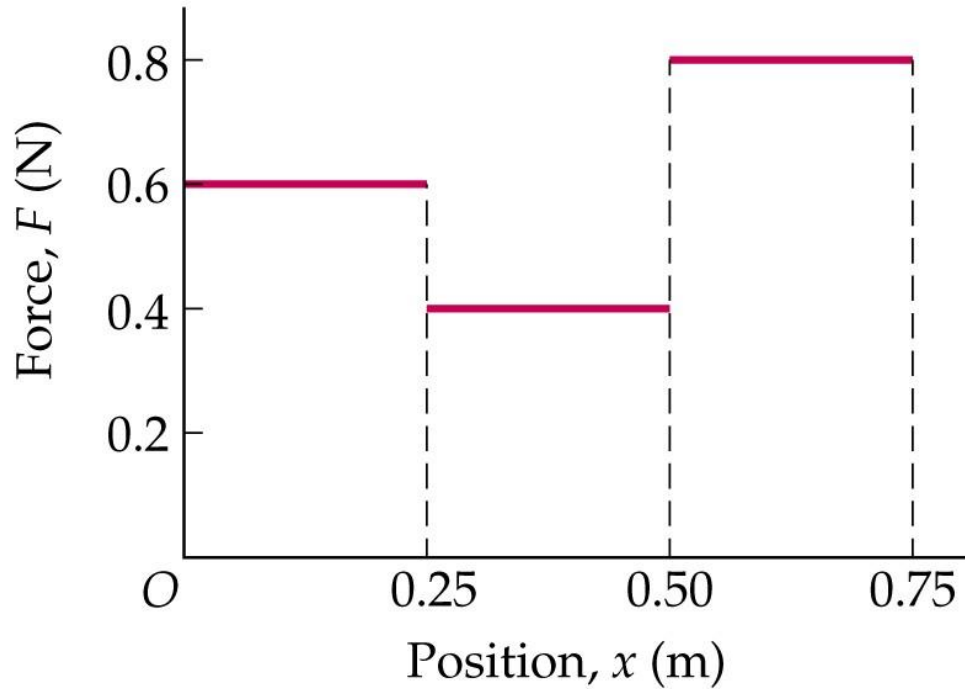


Constant force and work

- The force shown is a constant force.
- $W = F\Delta x$ can be used to calculate the work done by this force when it moves an object from x_a to x_b .
- The area under the curve from x_a to x_b can also be used to calculate the work done by the force

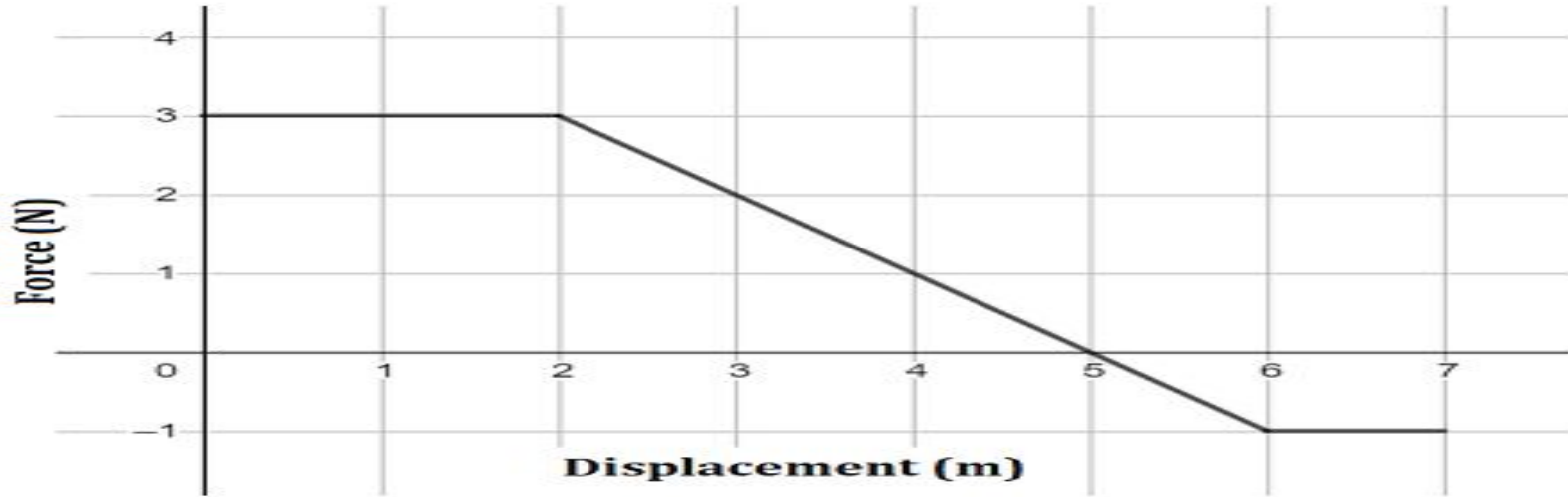


Sample Problem



- How much work is done by the force shown when it acts on an object and pushes it from $x = 0.25$ m to $x = 0.75$ m?

Sample Problem



- How much work is done by the force shown when it acts on an object and pushes it from $x = 2.0$ m to $x = 6.0$ m?

Kinetic Energy

- Energy due to motion
- $K = \frac{1}{2} m v^2$
 - K: Kinetic Energy in J
 - m: mass in kg
 - v: speed in m/s
- Unit: Joules
- Kinetic energy is independent of the direction of motion.

Sample problem

A 10.0 g bullet is moving right at a speed of 1.2 km/s.

a) What is the kinetic energy of the bullet?

7200 J

b) What is the bullet's kinetic energy if it is moving at the same speed 20° above the horizontal?

7200 J

c) What is the bullet's kinetic energy if it is moving at the same speed 60° south of west?

7200 J

Work and Energy

- ▣ *Work changes mechanical energy!*
- ▣ If an applied force does positive work on a system, it tries to increase mechanical energy.
- ▣ If an applied force does negative work, it tries to decrease mechanical energy.
- ▣ The two forms of mechanical energy are called potential and kinetic energy.

The Work-Energy Theorem

- The net work due to all forces equals the change in the kinetic energy of an object.
- $W_{\text{net}} = \Delta K$ *(when the earth or some massive object is not present)*
 - W_{net} : work due to the net of all forces acting on an object
 - ΔK : change in kinetic energy ($K_f - K_i$)

WORK-KINETIC ENERGY

- A spider monkey (mass 40 kg) is frolicking through the jungle at a speed of 3.0 m/s. All of a sudden, a jaguar appears and tries to eat the monkey. The monkey begins to run frantically — now he has a speed of 14.0 m/s. How much work does the monkey do to increase its speed?
- 3740 J

WORK-KINETIC ENERGY

- A 1000 kg car traveling at 30 m/s suddenly brakes and comes to a stop. If the wheels locked and friction applied a force of 5000 N, how far does the car travel?
- 90 m